

Md Bodrud-Doza, PhD

Postdoctoral Scholar | University of Guelph, Canada

Resilient Agricultural Landscapes and Sustainable Food Production

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Research Profile

Environmental scientist and geographer with over 10 years of combined research, teaching, and professional experience across academia, watershed management, NGOs, consulting, and international collaboration. My research focuses on agricultural landscapes, water quality, and spatial environmental decision-making. I integrate geospatial analysis, environmental monitoring, machine learning, and process-based modelling to identify environmental risks, prioritize interventions, and evaluate strategies for resilient and sustainable agricultural systems. My work builds on precision conservation, watershed management, groundwater and climate resilience, with a growing focus on Earth observation and GeoAI.

Education

PhD in Geography

University of Guelph, Canada | September 2022 - August 2026

Thesis: Advancing Precision Conservation in Tile-Drained Agricultural Watersheds through Integrated Geospatial Analysis, Machine Learning, and Process-Based Modeling.

Research focus: precision conservation, agricultural water quality, geospatial analysis, machine learning, and process-based watershed modelling.

MSc in Environmental Sciences

Jahangirnagar University, Bangladesh | 2014

Thesis: An integrated approach for groundwater pollution evaluation of Dhaka City using multivariate statistical analysis, groundwater pollution indices, geostatistical modeling, and human health risk assessment.

BSc (Honours) in Environmental Sciences

Jahangirnagar University, Bangladesh | 2010 - 2013

Project: Assessment of Groundwater Quality in Dhaka District Using GIS and Multivariate Statistical Techniques.

Academic and Research Appointments

Postdoctoral Scholar

Department of Geography, Environment and Geomatics, University of Guelph | September 2026 - Present

Developing a research program on resilient agricultural landscapes and sustainable food production, with emphasis on environmental decision support.

Expanding methodological capacity in Earth observation, GeoAI, environmental monitoring, geospatial analysis, machine learning, and process-based modelling.

Graduate Research Assistant

Department of Geography, Environment and Geomatics, University of Guelph | May 2024 - August 2026

Conducted distributed process-based watershed modelling to evaluate agricultural conservation practices in tile-drained landscapes.

Integrated GIS, statistical methods, machine learning, and ACPF-based planning to support field-scale conservation prioritization and Great Lakes water-quality management.

Collaborated with Environment and Climate Change Canada, Iowa State University, and academic partners.

Data Analyst

Ausable Bayfield Conservation Authority, Ontario | June 2024 - August 2024

Applied statistical and machine-learning approaches to long-term water-quality data from a Lake Huron tributary.

Supported watershed interpretation and development of monitoring and quality-assurance plans.

Research Program and Areas of Expertise

Agricultural Landscape Management: Precision and multifunctional conservation; field-level prioritization; nutrient-loss pathways; watershed and landscape resilience.

Geospatial and Data Science: GIS, geostatistics, spatial decision support, machine learning, spatial optimization, and interpretable environmental analytics.

Earth Observation and GeoAI - developing focus: Optical/radar remote sensing, Google Earth Engine, time-series analysis, and GeoAI for agricultural and environmental applications.

Water and Environmental Systems: Water quality, high-frequency monitoring, hydrology, nutrient transport, groundwater, and process-based environmental modelling.

Science-to-Decision Translation: Decision-support systems, policy engagement, monitoring strategy, stakeholder-oriented analysis, and research-to-practice partnerships.

Teaching Experience

Graduate Teaching Assistant

Department of Geography, Environment and Geomatics, University of Guelph | September 2022 - April 2024

GEOG*1350 Earth: Hazards and Global Change; GEOG*2000 Geomorphology; GEOG*3000 Fluvial Processes; GEOG*2480 Mapping and GIS; GEOG*3480 GIS and Spatial Analysis.

Led laboratories and seminars, supported student consultation, grading, rubric review, and exam invigilation, with hands-on instruction in geomorphology and GIS.

Teaching interests: GIS and geospatial analysis; remote sensing; GeoAI; environmental modelling; watershed science; agricultural sustainability; environmental decision support.

Research Funding, Grants, and Proposal Development

PhD project funding award from Environment and Climate Change Canada (2024-2026).

At BRAC, secured more than USD 10 million through 14 successful project proposals and co-designed additional projects intended to mobilize up to USD 100 million.

Co-led BRAC's accreditation to the Green Climate Fund as a Direct Access Entity and contributed to institutional climate-finance strategy.

At CARE Bangladesh, co-authored a successful ecosystem-based adaptation proposal that secured €35 million from the UK Foreign, Commonwealth & Development Office.

Coordinated and contributed to proposal development, resource mobilization, and partnership building across climate adaptation, agriculture, resilience, and sustainable-development programs.

Selected Awards and Recognition

Ontario Graduate Scholarship (OGS) (2025-2026). University of Guelph, Canada.

Cecil H. Franklin Graduate Scholarship in Soil & Water Conservation (2025). University of Guelph, Canada.

Ted Symons Scholarship (2025). University of Guelph, Canada.

Arthur D. Latornell Graduate Scholarship (2023), University of Guelph, Canada.

GEG Graduate Publication Award (2024, 2025). University of Guelph, Canada.

Three Minute Thesis (3MT) College Heat Winners (2023), College of Social and Applied Human Sciences, University of Guelph, Canada.

PhD Project funding award from Environment and Climate Change Canada (2024 to 2026).

International Doctoral Tuition Scholarship (IDTS), Special Dean's Award and Kiyoko Miyanishi Graduate Geography Scholarship for PhD program starting from Fall 2022 at the Department of Geography, Environment and Geomatics, University of Guelph, Ontario, Canada.

Erasmus Mundus Joint Master Degrees (EMJMD) scholarship (2022) for Master of Urban Climate & Sustainability (MURCS) (2022-2024) in Glasgow Caledonian University (UK), LAB University of Applied Sciences (Finland), University of Huelva (Spain), University of Applied Sciences Dresden (Germany). (Declined)

Chinese Government Scholarship (2017) for PhD degree in the Guangxi University, China. (Declined)

Peer-Reviewed Publications

Selected recent first-authored publications are listed first, followed by the complete publication record contained in the source CV.

First-Authored Articles

1. Bodrud-Doza, M., Yang, W., Liu, Y., Yerubandi, R., McKague, K., James, D., DeVries, B., & Fraser, E. D. G. (2026). Operationalizing precision conservation: Aligning best management practice prioritization with local needs in agricultural watersheds. *International Soil and Water Conservation Research*, 100688. <https://doi.org/10.1016/j.iswcr.2026.100688> Elsevier.

2. Bodrud-Doza, M., Yang, W., Liu, Y., Yerubandi, R., Thompson, M., McKague, K., James, D., DeVries, B., & Fraser, E. D. G. (2026). Advancing precision conservation through machine learning for field-level prioritization of agricultural best management practices. *International Soil and Water Conservation Research*, 100668. <https://doi.org/10.1016/j.iswcr.2026.100668> Elsevier.

3. Bodrud-Doza, M., Yang, W., Liu, Y., Yerubandi, R., Daggupati, P., DeVries, B., & Fraser, E. D. G. (2025). Evaluating best management practices for nutrient load reductions in tile-drained watersheds of the Laurentian Great Lakes Basin: A literature review. *Science of The Total Environment*, 965, 178657. <https://doi.org/10.1016/j.scitotenv.2025.178657> Elsevier.

4. Bodrud-Doza, M.; Yang, W.; Miranda, R Q; Martin, A; DeVries, B; Fraser, E D G. (2023). Towards implementing precision conservation practices in agricultural watersheds: A review of the use and prospects of Spatial Decision Support Tools and Systems. *Science of the Total Environment* 905C, 167118. <https://doi.org/10.1016/j.scitotenv.2023.167118> Elsevier.

5. Bodrud-Doza, M., Shammi, M., Bahlman, L., Islam, A.R.M.T., Rahman, M.M., 2020. Psychosocial and Socio-Economic Crisis in Bangladesh Due to COVID-19 Pandemic: A Perception-Based Assessment. *Front. Public Heal.* 8. <https://doi.org/10.3389/fpubh.2020.00341>. Frontier.

6. Bodrud-Doza, M., Islam, S.M.D.U., Rume, T., Quraishi, S.B., Rahman, M.S., Bhuiyan, M.A.H., 2020. Groundwater quality and human health risk assessment for safe and sustainable water supply of Dhaka City dwellers in Bangladesh. *Groundw. Sustain. Dev.* 10, 100374. <https://doi.org/10.1016/j.gsd.2020.100374>. Elsevier.
7. Bodrud-Doza, M., Bhuiyan, M.A.H., Islam, S.M.D.U., Quraishi, S.B., Muhib, M.I., Rakib, M.A., Rahman, M.S., 2019. Delineation of trace metals contamination in groundwater using geostatistical techniques: A study on Dhaka City of Bangladesh. *Groundw. Sustain. Dev.* 9. <https://doi.org/10.1016/j.gsd.2019.03.006>. Elsevier.
8. Bodrud-Doza, M., Bhuiyan, M.A.H., Islam, S.M.D.U., Rahman, M.S., Haque, M.M., Fatema, K.J., Ahmed, N., Rakib, M.A., Rahman, M.A., 2019. Hydrogeochemical investigation of groundwater in Dhaka City of Bangladesh using GIS and multivariate statistical techniques. *Groundw. Sustain. Dev.* 8, 226-244. <https://doi.org/10.1016/j.gsd.2018.11.008>. Elsevier.
9. Bodrud-Doza, M., Islam, S.M.D.U., Hasan, M.T., Alam, F., Haque, M.M., Rakib, M.A., Asad, M.A., Rahman, M.A., 2019. Groundwater pollution by trace metals and human health risk assessment in central west part of Bangladesh. *Groundw. Sustain. Dev.* 9, 100219. <https://doi.org/10.1016/j.gsd.2019.100219>. Elsevier.
10. Bodrud-Doza, M., Islam, A.R.M.T., Ahmed, F., Das, S., Saha, N., Rahman, M.S., 2016. Characterization of groundwater quality using water evaluation indices, multivariate statistics and geostatistics in central Bangladesh. *Water Sci.* 30, 19-40. <https://doi.org/10.1016/j.wsj.2016.05.001>. Taylor & Francis.

Co-Authored Articles

1. Zihad S.M.R.A., Islam A.R.M.T., Siddique M.A.B., Mia M.Y., Islam M.S., Islam M.A., Bari A.B.M.M., Bodrud-Doza M., Yakout S.M., Senapathi V., Chatterjee S. (2023). Fuzzy logic, geostatistics, and multiple linear models to evaluate irrigation metrics and their influencing factors in a drought-prone agricultural region. *Environ Res.* Volume 234, 116509. <https://doi.org/10.1016/j.envres.2023.116509>. Elsevier.
2. Ferdous, S., Chowdhury, F.N., Ali, M.L., Bodrud-Doza, M., Rahman, M.M. 2022. Assessment of urban sanitation status and management gaps in a metropolitan city, Bangladesh: Potential challenges to achieve SDG 6. *Front. Water* 4:950887. doi: 10.3389/frwa.2022.950887. *Frontier*.
3. Zaman, T., Tahsin, K.T., Rousseau, R.S., Kamal, A.B., Khan, M.R., Huq, S., Bodrud-Doza, M. 2022. An overview of disaster risk reduction and anticipatory action in Bangladesh. *Front. Clim.* 4:944736. doi: 10.3389/fclim.2022.944736. *Frontier*.
4. Rakib M. A., Quraishi S. B., Newaz M. A., Sultana J., Bodrud-Doza M., Rahman M. A., Patwary M. A., Bhuiyan M. A. H. 2022. Groundwater quality and human health risk assessment in selected coastal and floodplain areas of Bangladesh. *Journal of Contaminant Hydrology.* Volume 249, 104041. <https://doi.org/10.1016/j.jconhyd.2022.104041>. Elsevier.
5. Rahman, M.S., Parvez, S., Begum, B., Quraishi S., Choudhury T., Fatema K., Hosen M., Bodrud-Doza M., Rahman L., Sattar M.A 2022. Chemometric appraisal of water quality for domestic and agricultural purposes: a case study from establishing Rooppur Nuclear Power Plant (NPP) area, Pabna District, Bangladesh. *Environ Sci Pollut Res.* <https://doi.org/10.1007/s11356-022-19308-6>. Springer.
6. Sultan, M.B., Choudhury, T.R., Alam, M.N., Bodrud-Doza, M., Rahman, M.M. 2022. Soil, dust, and leaf-based novel multi-sample approach for urban heavy metal contamination appraisals in a megacity, Dhaka, Bangladesh. *Environmental Advances.* Volume 7, 100154. <https://doi.org/10.1016/j.envadv.2021.100154>. Elsevier.
7. Shammi, M., Rahmana, M.M., Ali, M.L., Khan, A.S.M., Siddique, M.A.B, Ashaduzzaman M., Bodrud-Doza M., Alam, G.M.M, Tareq, S.M. 2022. Application of short and rapid strategic environmental assessment (SEA) for biomedical waste management in Bangladesh. *Case Studies in Chemical and Environmental Engineering.* Volume 5, 100177. <https://doi.org/10.1016/j.cscee.2021.100177>. Elsevier.
8. Rahman, M.M., Bodrud-Doza, M., Shammi, M., Islam, A.R.M.T., Moniruzzaman Khan, A.S., 2021. COVID-19 pandemic, dengue epidemic, and climate change vulnerability in Bangladesh: Scenario assessment for strategic management and policy implications. *Environ. Res.* 192, 110303. <https://doi.org/10.1016/j.envres.2020.110303>. Elsevier.

9. Shammi, M., Bodrud-Doza, M., Islam, A.R.M.T., Rahman, M.M., 2021. Strategic assessment of COVID-19 pandemic in Bangladesh: comparative lockdown scenario analysis, public perception, and management for sustainability. *Environ. Dev. Sustain.* 23, 6148-6191. <https://doi.org/10.1007/s10668-020-00867-y>. Springer.
10. Bhuiyan, M.A.H., Bodrud-Doza, M., Rakib, M.A., Saha, B.B., Islam, S.M.D.U., 2021. Appraisal of pollution scenario, sources and public health risk of harmful metals in mine water of Barapukuria coal mine industry in Bangladesh. *Environ. Sci. Pollut. Res.* 28, 22105-22122. <https://doi.org/10.1007/s11356-020-11999-z>. Springer.
11. Hasan, S., Islam, M.A., Bodrud-Doza, M., 2021. Crisis perception and consumption pattern during COVID-19: do demographic factors make differences? *Heliyon* 7, e07141. <https://doi.org/10.1016/j.heliyon.2021.e07141>. Elsevier.
12. Hassan, M.S., Bhuiyan, M.A.H., Tareq, F., Bodrud-Doza, M., Tanu, S.M., Rabbani, K.A., 2021. Relationship between COVID-19 infection rates and air pollution, geo-meteorological, and social parameters. *Environ. Monit. Assess.* 193, 29. <https://doi.org/10.1007/s10661-020-08810-4>. Springer.
13. Islam, A.R.M.T., Hasanuzzaman, M., Shammi, M., Salam, R., Bodrud-Doza, M., Rahman, M.M., Mannan, M.A., Huq, S., 2021. Are meteorological factors enhancing COVID-19 transmission in Bangladesh? Novel findings from a compound Poisson generalized linear modeling approach. *Environ. Sci. Pollut. Res.* 28, 11245-11258. <https://doi.org/10.1007/s11356-020-11273-2>. Springer.
14. Islam, S.M.D.U., Mondal, P.K., Ojong, N., Bodrud-Doza, M., Siddique, M.A.B., Hossain, M., Mamun, M.A., 2021. Water, sanitation, hygiene and waste disposal practices as COVID-19 response strategy: insights from Bangladesh. *Environ. Dev. Sustain.* 23, 11953-11974. <https://doi.org/10.1007/s10668-020-01151-9>. Springer.
15. Mamun, M.A., Sakib, N., Gozal, D., Bhuiyan, A.I., Hossain, S., Bodrud-Doza, M., Al Mamun, F., Hosen, I., Safiq, M.B., Abdullah, A.H., Sarker, M.A., Rayhan, I., Sikder, M.T., Muhit, M., Lin, C.-Y., Griffiths, M.D., Pakpour, A.H., 2021. The COVID-19 pandemic and serious psychological consequences in Bangladesh: A population-based nationwide study. *J. Affect. Disord.* 279, 462-472. <https://doi.org/10.1016/j.jad.2020.10.036>. Elsevier.
16. Siddique, A.B., Islam, A.R.M.T., Hossain, M.S., Khan, R., Akbor, M. A., Hasanuzzaman, M., Sajid, M.W.M, Mia, M.Y., Mallick, J., Rahman, M.S., Rahman, M.M, Bodrud-Doza, M. 2021. Multivariate statistics and entropy theory for irrigation water quality and entropy-weighted index development in a subtropical urban river, Bangladesh. *Environ Sci Pollut Res.* <https://doi.org/10.1007/s11356-021-16343-7>. Springer.
17. Islam, A.R.M.T., Kabir, M.M., Faruk, S., Al Jahin, J., Bodrud-Doza, M., Didar-ul-Alam, M., Bahadur, N.M., Mohinuzzaman, M., Fatema, K.J., Safiur Rahman, M., Choudhury, T.R., 2021. Sustainable groundwater quality in southeast coastal Bangladesh: co-dispersions, sources, and probabilistic health risk assessment. *Environ. Dev. Sustain.* <https://doi.org/10.1007/s10668-021-01447-4>. Springer.
18. Rakib, M.A., Sasaki, J., Matsuda, H., Quraishi, S.B., Mahmud, M.J., Bodrud-Doza, M., Ullah, A.K.M.A., Fatema, K.J., Newaz, M.A., Bhuiyan, M.A.H., 2020. Groundwater salinization and associated co-contamination risk increase severe drinking water vulnerabilities in the southwestern coast of Bangladesh. *Chemosphere* 246, 125646. <https://doi.org/10.1016/j.chemosphere.2019.125646>. Elsevier.
19. Islam, S.M.D., Bodrud-Doza, M., Khan, R.M., Haque, M.A., Mamun, M.A., 2020. Exploring COVID-19 stress and its factors in Bangladesh: A perception-based study. *Heliyon* 6, e04399. <https://doi.org/10.1016/j.heliyon.2020.e04399>. Elsevier.
20. Islam, S.M.D., Safiq, M.B., Bodrud-Doza, M., Mamun, M.A., 2020. Perception and Attitudes Toward PPE-Related Waste Disposal Amid COVID-19 in Bangladesh: An Exploratory Study. *Front. Public Heal.* 8. <https://doi.org/10.3389/fpubh.2020.592345>. Frontier.
21. Mamun, M.A., Bodrud-Doza, M., Griffiths, M.D., 2020. Hospital suicide due to non-treatment by healthcare staff fearing COVID-19 infection in Bangladesh? *Asian J. Psychiatr.* 54, 102295. <https://doi.org/10.1016/j.ajp.2020.102295>. Elsevier.
22. Rahman, Mostafizur M.M., Bodrud-Doza, M., Griffiths, M.D., Mamun, M.A., 2020. Biomedical waste amid COVID-19: perspectives from Bangladesh. *Lancet Glob. Heal.* 8, e1262. [https://doi.org/10.1016/S2214-109X\(20\)30349-1](https://doi.org/10.1016/S2214-109X(20)30349-1). Elsevier.

23. Shammi, M., Bodrud-Doza, M., Islam, A.R.M.T., Rahman, M.M., 2020. COVID-19 pandemic, socioeconomic crisis and human stress in resource-limited settings: A case from Bangladesh. *Heliyon* 6, e04063. <https://doi.org/10.1016/j.heliyon.2020.e04063>. Elsevier.
24. Rahman, M.M., Bodrud-Doza, M., Siddiqua, M.T., Zahid, A., Islam, A.R.M.T., 2020. Spatiotemporal distribution of fluoride in drinking water and associated probabilistic human health risk appraisal in the coastal region, Bangladesh. *Sci. Total Environ.* 724, 138316. <https://doi.org/10.1016/j.scitotenv.2020.138316>. Elsevier.
25. Saha, N., Bodrud-Doza, M., Islam, A.R.M.T., Begum, B.A., Rahman, M.S., 2020. Hydrogeochemical evolution of shallow and deeper aquifers in central Bangladesh: arsenic mobilization process and health risk implications from the potable use of groundwater. *Environ. Earth Sci.* 79, 1-18. <https://doi.org/10.1007/s12665-020-09228-4>. Springer.
26. Habib, M.A., Islam, A.R.M.T., Bodrud-Doza, M., Mukta, F.A., Khan, R., Bakar Siddique, M.A., Phoungthong, K., Techato, K., 2020. Simultaneous appraisals of pathway and probable health risk associated with trace metals contamination in groundwater from Barapukuria coal basin, Bangladesh. *Chemosphere* 242, 125183. <https://doi.org/10.1016/j.chemosphere.2019.125183>. Elsevier.
27. Islam, A.R.M.T., Islam, H.M.T., Mia, M.U., Khan, R., Habib, M.A., Bodrud-Doza, M., Siddique, M.A.B., Chu, R., 2020. Co-distribution, possible origins, status and potential health risk of trace elements in surface water sources from six major river basins, Bangladesh. *Chemosphere* 249. <https://doi.org/10.1016/j.chemosphere.2020.126180>. Elsevier.
28. Shammi, M., Sultana, A., Hasan, N., Mostafizur Rahman, M., Saiful Islam, M., Bodrud-Doza, M., Khabir Uddin, M., 2020. Pesticide exposures towards health and environmental hazard in Bangladesh: A case study on farmers' perception. *J. Saudi Soc. Agric. Sci.* 19, 161-173. <https://doi.org/10.1016/j.jssas.2018.08.005>. Elsevier.
29. Bhuiyan, M.A.H., Chandra Karmaker, S., Bodrud-Doza, M., Rakib, M.A., Saha, B.B., 2021. Enrichment, sources and ecological risk mapping of heavy metals in agricultural soils of dhaka district employing SOM, PMF and GIS methods. *Chemosphere* 263, 128339. <https://doi.org/10.1016/j.chemosphere.2020.128339>. Elsevier.
30. Akbor, M.A., Rahman, M.M., Bodrud-Doza, M., Haque, M.M., Siddique, M.A.B., Ahsan, M.A., Bondad, S.E.C., Uddin, M.K., 2020. Metal pollution in water and sediment of the Buriganga river, Bangladesh: An ecological risk perspective. *Desalin. Water Treat.* 193, 284-301. <https://doi.org/10.5004/dwt.2020.25805>. Elsevier.
31. Khan, R., Islam, M.S., Tareq, A.R.M., Naher, K., Islam, A.R.M.T., Habib, M.A., Siddique, M.A.B., Islam, M.A., Das, S., Rashid, M.B., Ullah, A.K.M.A., Miah, M.M.H., Masrura, S.U., Bodrud-Doza, M., Sarker, M.R., Badruzzaman, A.B.M., 2020. Distribution, sources and ecological risk of trace elements and polycyclic aromatic hydrocarbons in sediments from a polluted urban river in central Bangladesh. *Environ. Nanotechnology, Monit. Manag.* 14, 100318. <https://doi.org/10.1016/j.enmm.2020.100318>. Elsevier.
32. Azam, G., Huda, M.E., Bhuiyan, M.A.H., Mohinuzzaman, M., Bodrud-Doza, M., Islam, S.M.D.-U., 2019. Climate Change and Natural Hazards Vulnerability of Char Land (Bar Land) Communities of Bangladesh: Application of the Livelihood Vulnerability Index (LVI). *Glob. Soc. Welf.* 8, 93-105. <https://doi.org/10.1007/s40609-019-00148-1>. Springer.
33. Shammi, M., Rahman, M., Bondad, S., Bodrud-Doza, M., 2019. Impacts of Salinity Intrusion in Community Health: A Review of Experiences on Drinking Water Sodium from Coastal Areas of Bangladesh. *Healthcare* 7, 50. <https://doi.org/10.3390/healthcare7010050>. MDPI.
34. Shammi, M., Rahman, M.M., Islam, M.A., Bodrud-Doza, M., Zahid, A., Akter, Y., Quaiyum, S., Kurasaki, M., 2017. Spatio-temporal assessment and trend analysis of surface water salinity in the coastal region of Bangladesh. *Environ. Sci. Pollut. Res.* 24, 14273-14290. <https://doi.org/10.1007/s11356-017-8976-7>. Springer.
35. Rakib, M.A., Sasaki, J., Pal, S., Newaz, M.A., Bodrud-Doza, M., Bhuiyan, M.A.H., 2019. An investigation of coastal vulnerability and internal consistency of local perceptions under climate change risk in the southwest part of Bangladesh. *J. Environ. Manage.* 231, 419-428. <https://doi.org/10.1016/j.jenvman.2018.10.054>. Elsevier.
36. Ahmed, N., Bodrud-Doza, M., Islam, A.R.M.T., Hossain, S., Moniruzzaman, M., Deb, N., Bhuiyan, M.A.Q., 2019. Appraising spatial variations of As, Fe, Mn and NO₃ contaminations associated health risks of drinking water from Surma basin, Bangladesh. *Chemosphere* 218, 726-740. <https://doi.org/10.1016/j.chemosphere.2018.11.104>. Elsevier.

37. Ahmed, N., Bodrud-Doza, M., Islam, S.M.D.-U., Choudhry, M.A., Muhib, M.I., Zahid, A., Hossain, S., Moniruzzaman, M., Deb, N., Bhuiyan, M.A.Q., 2019. Hydrogeochemical evaluation and statistical analysis of groundwater of Sylhet, north-eastern Bangladesh. *Acta Geochim.* 38, 440-455. <https://doi.org/10.1007/s11631-018-0303-6>. Elsevier.
38. Islam, A.R.M.T., Bodrud-Doza, M., Rahman, M.S., Amin, S.B., Chu, R., Al Mamun, H., 2019. Sources of trace elements identification in drinking water of Rangpur district, Bangladesh and their potential health risk following multivariate techniques and Monte-Carlo simulation. *Groundw. Sustain. Dev.* 9, 100275. <https://doi.org/10.1016/j.gsd.2019.100275>. Elsevier.
39. Rahman, M.M., Islam, M.A., Bodrud-Doza, M., Muhib, M.I., Zahid, A., Shammi, M., Tareq, S.M., Kurasaki, M., 2018. Spatio-Temporal Assessment of Groundwater Quality and Human Health Risk: A Case Study in Gopalganj, Bangladesh. *Expo. Heal.* 10, 167-188. <https://doi.org/10.1007/s12403-017-0253-y>. Springer.
40. Islam, M.A., Rahman, M.M., Bodrud-Doza, M., Muhib, M.I., Shammi, M., Zahid, A., Akter, Y., Kurasaki, M., 2018. A study of groundwater irrigation water quality in south-central Bangladesh: a geo-statistical model approach using GIS and multivariate statistics. *Acta Geochim.* 37, 193-214. <https://doi.org/10.1007/s11631-017-0201-3>. Elsevier.
41. Rakib, M.A., Islam, S., Nikolaos, I., Bodrud-Doza, M., Bhuiyan, M.A.H., 2017. Flood vulnerability, local perception and gender role judgment using multivariate analysis: A problem-based "participatory action to Future Skill Management" to cope with flood impacts. *Weather Clim. Extrem.* 18, 29-43. <https://doi.org/10.1016/j.wace.2017.10.002>. Elsevier.
42. Islam, A.R.M.T., Shen, S.H., Bodrud-Doza, M., 2017. Assessment of arsenic health risk and source apportionment of groundwater pollutants using multivariate statistical techniques in Chapai-Nawabganj district, Bangladesh. *J. Geol. Soc. India* 90, 239-248. <https://doi.org/10.1007/s12594-017-0705-9>. Springer.
43. Islam, A.R.M.T., Shen, S., Haque, M.A., Bodrud-Doza, M., Maw, K.W., Habib, M.A., 2018. Assessing groundwater quality and its sustainability in Joypurhat district of Bangladesh using GIS and multivariate statistical approaches. *Environ. Dev. Sustain.* 20, 1935-1959. <https://doi.org/10.1007/s10668-017-9971-3>. Springer.
44. Atikul Islam, M., Zahid, A., Rahman, M.M., Rahman, M.S., Islam, M.J., Akter, Y., Shammi, M., Bodrud-Doza, M., Roy, B., 2017. Investigation of Groundwater Quality and Its Suitability for Drinking and Agricultural Use in the South Central Part of the Coastal Region in Bangladesh. *Expo. Heal.* 9, 27-41. <https://doi.org/10.1007/s12403-016-0220-z>. Springer.
45. Islam, A.R.M.T., Ahmed, N., Bodrud-Doza, M., Chu, R., 2017. Characterizing groundwater quality ranks for drinking purposes in Sylhet district, Bangladesh, using entropy method, spatial autocorrelation index, and geostatistics. *Environ. Sci. Pollut. Res.* 24, 26350-26374. <https://doi.org/10.1007/s11356-017-0254-1>. Springer.
46. Islam, A.R.M.T., Shen, S., Bodrud-Doza, M., Rahman, M.S., 2017. Assessing irrigation water quality in Faridpur district of Bangladesh using several indices and statistical approaches. *Arab J Geosci* 10, 418. <https://doi.org/10.1007/s12517-017-3199-2>. Springer.
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50. Shammi, M., Rahman, R., Rahman, M.M., Moniruzzaman, M., Bodrud-Doza, M., Karmakar, B., Uddin, M.K., 2016. Assessment of salinity hazard in existing water resources for irrigation and potentiality of conjunctive uses: a case report from Gopalganj District, Bangladesh. *Sustain. Water Resour. Manag.* 2, 369-378. <https://doi.org/10.1007/s40899-016-0064-5>. Springer.

Book Chapters

1. Sifa, S.F., Sultana, R., Bodrud-Doza, M., 2021. Climate Change and COVID-19: Crisis Within Crises for Eradication of Poverty in Bangladesh. Muthu S.S (Eds), COVID19: Environmental Sustainability and Sustainable Development Goals (pp. 169-188). https://doi.org/10.1007/978-981-16-3860-2_7. Springer.
2. Ali, M.M., Afrin, S., Siddique, A.B., Bodrud-Doza, M., Rahman M.M. (2022). Chapter 8 - Metal contamination in groundwater of Bangladesh. Tiwari, A.K., Kumar, A., Singh B.K., Singh, T.N., Suozzi, N., Matta, G., Russo, S.L, (Eds.) Water Scarcity, Contamination and Management (Volume 5, pp 109-131). <https://doi.org/10.1016/B978-0-323-85378-1.00008-8>. Elsevier.

Working Papers

1. Coger, T., A. Dinshaw, S. Tye, B. Kratzer, M. Thazin Aung, E. Cunningham, C. Ramkissoon, S. Gupta, Bodrud-Doza M., A. Karamallis, S. Mbewe, A. Granderson, G. Dolcemascolo, A. Tewary, A. Mirza, and A. Carthy. 2022. "Locally Led Adaptation: From Principles to Practice." Working Paper. Washington, DC: World Resources Institute (WRI). Available online at <https://doi.org/10.46830/wriwp.21.00142>.
2. Steinbach, D., Bahadur, A., Shakya, C., Thazin Aung, M., Burton, C., Gallagher, C., Mbewe, S., Greene, S., Regmi, B., Granderson, A., Ramkissoon, C., Kostka, W., Andon, L., Greenstone-Alefaio, T., Dolcemascolo, G., Gupta, S., Tewary, A., Lopez, M., Barnes, J., Mirza, A., Bodrud-Doza, M., Akhter, F., Rozario, S., Reyes, C. (2022). The good climate finance guide for investing in locally led adaptation. IIED, London. <https://www.iied.org/21231iied>

Conference Abstracts

1. Bodrud-Doza, M. (2025). Towards implementing precision conservation practices in agricultural watersheds: A review of the use and prospects of Spatial Decision Support Tools and Systems. 1st International Conference on Canadian Agri-food & Rural Advisory, Extension, and Education (CAREE). Guelph, Ontario, Canada.
2. Bodrud-Doza M., Rahman S., Bhuiyan M.A.H., Quraishi S. B., Choudhury T. R., Fatema K. J., Ji H. G. (2018). Health risk assessment of groundwater from shallow, intermediate and deep aquifer of Dhaka City, Bangladesh. 39th Annual Conference of Bangladesh Chemical Society. Bangladesh Chemical Congress-2018.
3. Bodrud-Doza M., Chowdhury N., Rahman M. S. (2017). Risk assessment of trihalomethanes (THMs) formation in the supplied water of Dhaka city, Bangladesh. Abstract Book. Symposium on Environmental Chemistry for Securing Water Quality. Bangladesh Journal of Scientific and Industrial Research. 52 (Special Issue).
4. Asad M. A., Bodrud-Doza M., Das M. (2018). Agricultural adaptation strategies to climate change - induced salinity intrusion in Southwest coastal Bangladesh. International Conference on Global Warming and Climate Change 2018. International Institute of Knowledge Management. 04th - 05th October 2018 in Bangkok, Thailand.
5. Rakib M.A., Islam S., Nikolaos I., Bodrud-Doza M., Bhuiyan M. A. H. (2018). Flood vulnerability, local perception and gender role judgment using multivariate analysis. The 4th Gobeshona Conference for Climate Research in Bangladesh. 08-11 January, 2018 Dhaka, Bangladesh.
6. Bodrud-Doza M., Jolly Y. N., Begum B. A., Fatema K. J., Bhuiyan M. A. H., Quraishi S. B., Rahman M. S. (2017). An integrated approach for groundwater pollution evaluation of Dhaka City using multivariate statistical analysis, groundwater pollution indices, geostatistical modeling and human health risk assessment. National Conference on Physics-2017. Bangladesh Physical Society. Atomic Energy Centre, Dhaka.
7. Anny F. A., Kabir M. M., Bodrud-Doza M. (2015). Determination of surface water quality in Savar urban area, Dhaka, Bangladesh. Annual Botanical Conference-2015. Bangladesh Botanical Society.

Technical Reports and Policy Contributions

1. Bodruddoza, M., Burton, E., Dahunsi, J. E., Livingston, J., Mengistu-Zelleke, N., & Pollet, S. (2023). Upstream evidence report 2023. Arrell Food Institute & Ontario Agricultural College, University of Guelph. Available at: <https://arrellfoodinstitute.ca/net-zero-challenge/>
2. Farhad S. M., Hasan M., Rahman M. S., Rahman H. M. H., Raju M. R. H., Bodrud-Doza M., Sarkar S. (2022). Low-cost climate resilient urban housing: BRAC's experiences. Published by BRAC. 47 pages.
3. Khan A. S. M., Bodrud-Doza M., Khan R. M., Siddique M. A. B., Ashadudzaman M. (2020). Mainstreaming Climate Change at BRAC. Published by BRAC. 64 pages.
4. Nasir N., Rahman M. F., Bodrud-Doza M., Khan A. S. M., (2020). Climate Resilience Framework of BRAC. Published by BRAC. 48 pages.
5. Huq S., Haque S., Bodrud-Doza M., Nasir N., Yusuf M., Huq S. (2018). Assessment of Socio-economic Impacts of Cross-Border Electricity Trade (CBET) in Bangladesh. Published by Integrated Research and Action for Development (IRADe), India. 52 pages.
6. Schulenburg A., Kamruzzaman M., Bodrud-Doza M., Mahmud M., Khan R. A., Haque M.A., Islam N., Talukder M. B. A., Auvi P., Rahman M. F. (2017). Adaptation Technology in Bangladesh. Jointly published by CCDB, ICCCAD and Practical Action. 73 pages.

Policy Brief

Huq S., Bodrud-Doza M., Tahsin K. T., Khan M. R. (2022). What does the IPCC Working Group II say about Bangladesh in its Sixth Assessment Report?. Policy Brief. Dhaka, Bangladesh: ICCCAD. Available online at <https://www.icccad.net/publication/s/policy-brief/what-does-the-ipcc-working-group-ii-say-about-bangladesh-in-its-sixth-assessment-report/>

Selected Public and Policy Writing

1. Cogger T., Tye S., Karamallis A., Mirza A., Bodrud-Doza M., Cunningham E.M. (01 November 2022). Getting Locally Led Adaptation Right: Examples from Around the World. <https://www.wri.org/insights/getting-locally-led-adaptation-right-examples-around-world>
2. Bodrud-Doza M., Tahsin K. T., (11 October, 2022). The ways to reinvent climate change adaptation in Bangladesh and beyond. <https://www.tbsnews.net/thoughts/ways-reinvent-climate-change-adaptation-bangladesh-and-beyond-511610>
3. Huq S., Rozario S. R., Bodrud-Doza M. (18 August 2022). Are Bangladesh's development measures leading to climate change readiness or maladaptation?. <https://news.mongabay.com/2022/08/are-bangladeshs-development-measures-leading-to-climate-change-readiness-or-maladaptation-commentary/>
4. Bodrud-Doza M., Mahfuz M. (18 July 2022). What local NGOs must do to secure climate finance. <https://www.thedailystar.net/opinion/views/news/what-local-ngos-must-do-secure-climate-finance-3073301>
5. Tahsin K. T., Bodrud-Doza M. (July 2022). Effective adaptation for avoiding maladaptation. <https://www.dhakatribune.com/tribune-climate/2022/08/25/effective-adaptation-for-avoiding-maladaptation>
6. Bodrud-Doza M. (10 June 2022). Boosting disaster preparedness in the Ganges Delta. D+C Development and Cooperation. <https://www.dandc.eu/en/article/bangladesh-has-made-remarkable-progress-towards-climate-resilience>
7. Mahfuz M., Bodrud-Doza M. (9 May 2022). It's time to popularise climate resilient housing. <https://www.thedailystar.net/views/opinion/news/its-time-popularise-climate-resilient-housing-3019336>
8. Bodrud-Doza M., Khan S. (24 April 2022). Addressing food security and climate change through regenerative agriculture (Link: <https://www.tbsnews.net/thoughts/addressing-food-security-and-climate-change-through-regenerative-agriculture-408578>)

9. Huq S., Bodrud-Doza M., Tahsin K. T. (2022). Resilience and prosperity under a changing climate. (Link: <https://www.icccad.net/publications/journal-article/resilience-and-prosperity-under-a-changing-climate/>)
10. Huq S., Bodrud-Doza M., Tahsin K. T., Khan M. R. (20 March 2022). What does the IPCC say about Bangladesh in its Sixth Assessment Report of Working Group II? (Link: <https://www.tbsnews.net/thoughts/what-does-ipcc-say-about-bangladesh-its-sixth-assessment-report-working-group-ii-387822>)

Professional and Leadership Experience

Operation and Business Development Manager

International Centre for Climate Change and Development (ICCCAD), Independent University, Bangladesh | January - August 2022

Coordinated ICCCAD's research portfolio and Locally Led Adaptation Program; oversaw project implementation, resource mobilization, partnerships, and capacity development.

Guided research papers, policy briefs, project reports, stakeholder training, and climate-resilience proposal development.

Consultant - Project Proposal Writer

CARE Bangladesh | August - November 2022

Co-developed climate-resilience and ecosystem-based adaptation proposals with multidisciplinary teams.

Co-authored a successful ecosystem-based adaptation proposal that secured €35 million from FCDO.

Consultant

Ispahani Agro Limited | March - August 2022

Conducted research and assessments for climate-resilient seeds, bio-pesticides, sustainable agricultural investment, ESG, financial inclusion, and smallholder digital services.

Manager, Climate Resilience and Development

BRAC Climate Change Program | April 2018 - January 2022

Led project development, proposal design, resource mobilization, climate-adaptation implementation, environmental and social safeguards, and climate-smart agriculture initiatives.

Built partnerships with UNDP, government agencies, and international donors; co-developed institutional climate-resilience frameworks and knowledge products.

Research Officer

International Centre for Climate Change and Development (ICCCAD) | October 2016 - March 2018

Conducted climate-adaptation and resilience research, GIS analysis, technical writing, policy research, proposal development, and capacity building.

Junior Consultant

International Union for Conservation of Nature (IUCN), Bangladesh | October - December 2016

Conducted GIS analysis and mapping to support wildlife conservation planning and Vulture Safe Zone initiatives.

Academic and Professional Service

Journal peer review across environmental science and water-resource journals, including Journal of Environmental Management, Environmental Research, Journal of Cleaner Production, Journal of Hydrology, Water Research, Agricultural Water Management, and Journal of Hazardous Materials.

Committee Member, Arrell Food Chair in Food and Society Search Committee, University of Guelph (2023); represented graduate students in faculty hiring review and interviews.

Member, Canadian Water Resources Association (2025-present)

Member, Canadian Geophysical Union (2024-present)

Member, Canadian Geomorphology Research Group (2024-present)

Technical Skills

Programming & Data Science: Python, R, JavaScript, SQL; statistical and machine learning modeling for environmental and geospatial applications.

GIS & Remote Sensing: ArcGIS, QGIS, Google Earth Engine, ERDAS IMAGINE, WhiteboxTools, RockWorks, Surfer.

Hydrologic & Environmental Modeling: SWAT, IMWEBs, ACPF, Aquachem; scenario analysis for water quality, BMP effectiveness, and climate resilience.

Data Analytics & Visualization: SPSS, DeltaGraph, Tableau; advanced geospatial/statistical analysis and interactive data dashboards.

Field Data Collection: Mobile-based tools (ODK Collect, KoBoToolbox, Google Forms) for real-time surveys and environmental monitoring.

AI & Digital Tools: Experience with AI-assisted research and productivity platforms (ChatGPT, Gemini, Microsoft Copilot, Claude).

Communication & Media: Infographic design (Adobe Illustrator, Photoshop), technical writing, and presentation development.

Multimedia Production: Video documentation and editing (Adobe Premiere Pro, After Effects) for research communication and training.

Teaching & Training: Designing and delivering capacity-building sessions in GIS, modeling, climate adaptation, and environmental management.

Languages: Fluent in English and Bangla (reading, writing, speaking, listening).

Selected Professional Training

Spatial Data Science: The New Frontier in Analytics - Esri (2024)

GIS for Climate Action - Esri (2024)

Cartography - Esri (2024)

Creating Python Scripts for Raster Analysis - Esri (2024)

Python for Everyone - Esri (2024)

Remote Sensing Image Acquisition, Analysis and Applications - Coursera (2023)

Spatial Analysis and Satellite Imagery in a GIS - Coursera (2023)

SWAT CUP Calibration & Validation with ArcSWAT - Udemy (2024)

ArcSWAT Model with ArcGIS - Run for Any Study Area - Udemy (2022)

Crash Course on Python - Google (2024)

Carbon Sequestration in Watersheds - Holland College (2024)

Climate Smart Agriculture: Fundamentals; Data Management & Information Systems - Holland College (2024)

Using LiDAR to Understand Ecosystems - Holland College (2024)

Agricultural Conservation Planning Framework (ACPF) Technical Training - Iowa State University & USDA ACPF Hub (2022)

Professional Profiles

Google Scholar: <https://scholar.google.com/citations?user=1VymNoIAAAAJ&hl=en>

ORCID: <https://orcid.org/0000-0003-4694-4077>

LinkedIn: <https://www.linkedin.com/in/md-bodrud-doza-33156a80/>

GitHub: <https://github.com/bdzion>

ResearchGate: <https://www.researchgate.net/profile/Md-Bodrud-Doza>

ScienceDirect: <https://www.sciencedirect.com/author/57189758827/m-bodrud-doza>